CAD VR: Guarded Evidence Control for Fast and Slow Agentic Video Retrieval

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Abstract. Video event retrieval, as posed by the Ho Chi Minh City AI Challenge 2026 (AIC 2026), requires locating precise moments from multimodal descriptions under time constraints. Embedding scores and agent confidence alone do not establish whether the evidence is sufficient to stop searching. We introduce CAD VR, the system of team APCS ⊕ CNTN, which couples fast multimodal retrieval with slower tool-using agents through a provenance-aware evidence board, typed verification and action selection, and a code-level stop guard. We evaluate six configurations on 86 AIC 2026 TEST queries and conduct a subsequent live ablation of final verification. In that ablation, CAD VR achieves 58.1% strict R@1, versus 47.7% for always-two agents and 50.0% for always-two agents with final verification; its advantage over the latter remains inconclusive (exact McNemar $p = 0.092$). Relative to always-two agents, median latency increases from 53.7 to 75.8 s with negligible agent-call reduction. In the AIC 2026 final round, team APCS ⊕ CNTN was rated Outstanding overall. These findings show how to make retrieval decisions auditable, while the accuracy–latency trade-off motivates improved stopping policies and manual–agent cooperation.

Keywords: Video event retrieval · LLM agents · Fast and slow control · Decision models · Evidence verification · Multimodal retrieval

1 Introduction

The Ho Chi Minh City AI Challenge 2026 (AIC 2026) [12] poses tasks that map descriptions to a known moment (KIS), a moment and answer (QA), or an ordered event chain (TRAKE). Joint embeddings generate candidates quickly [5, 29], but exact localisation remains difficult: queries combine actions, appearance, text and speech, while broadcasts repeat similar scenes. On our benchmark, retrieval ranks the correct video first on 45.3% of queries, yet strict R@1 is only 15.1%.

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Tool-using agents search, inspect frames and refine queries [25, 28, 40, 43]. Running both agents costs tens of seconds; their confidence is uncalibrated, and disagreement alone does not justify further search.

CAD VR, built by team APCS ⊕ CNTN for AIC 2026, makes escalation, evidence assessment and stopping auditable through fast and slow control [21,27]: System-1 retrieves, a non-generative decision model [37] assesses evidence and selects typed actions, and System-2 agents search and inspect images. "Fast and slow" denotes these roles, not lower latency. Our contributions are:

- **Typed evidence-guided control**: a bounded controller separates candidate generation from two soft-role agents sharing ten Model Context Protocol (MCP) tools [1] in an operator-facing multimodal retrieval system.
- **Evidence board and stop guard**: provenance, typed assessments and code rules expose abstention, escalation and budget stops in auditable traces.
- **Controlled evaluation and diagnostics**: a live C/C+V/E ablation and same-evidence pre/post verification contrasts within C+V distinguish final-ranking effects from the broader controller package. Independent hint prefixes and group-level metrics complement this component analysis.
- **Final-round report**: official AIC 2026 results, cases and operator practices.

2 Related Work

Interactive video retrieval. VBS systems combine embeddings and browsing [29, 34], spatial and temporal queries [8, 20, 22], answer grounding [36], fusion [23] and graded autonomy [16]. AIC systems use dynamic programming and visual verification [15, 31]; agents mediate video and lifelog search [17, 19, 38]. CAD VR adds typed control and an evidence guard above search agents.

LLM agents for video. Agents inspect retrieved frames [40], select tools [43], separate planning, grounding and verification [28], plan temporal scope [25] and coordinate retrieval [41]. We separate this interleaved reasoning and acting [42] from typed control and code-level stopping.

Fast and slow control and routing. SwiftSage pairs fast policies with slow planning [27]; FrugalGPT and RouteLLM trade model cost for quality [7, 32]. Unlike free-form judge rationales [45], CAD VR uses typed decision probabilities and a separate guard, with calibration diagnostics [14].

3 System Overview

The system (Fig. 1) indexes 1,478 broadcasts plus batch-two news, 298 traffic-camera and 12 cycling recordings, with organiser (382,299 frames) and InfoShot++ (2,212,383 frames) profiles. Our InfoShot++ sampling algorithm extends InfoShot [44] with multiple unusual within-shot candidates, native-frame-rate refinement around text/motion changes, and temporal gap filling. Its benefit is not

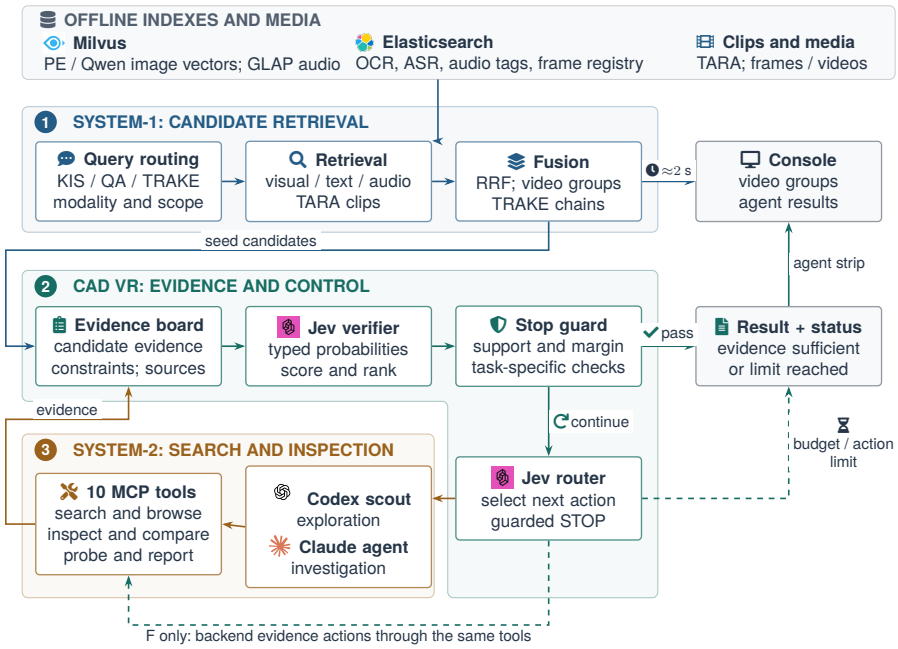


Fig. 1. CAD VR architecture: retrieval, evidence verification, guarded stopping and agent/tool actions. Dashed paths denote F-only macros and limit termination; evidence returns to the query-local board.

ablated. Milvus [39] stores PE-Core [5], Qwen3-VL [26] and GLAP [11] vectors; Elasticsearch stores OCR [18], WhisperX-aligned Vietnamese ASR [4, 24], audio tags [10] and frame/time mappings. TARA [3] indexes 496,155 clips across core and batch-two videos (partial cycling coverage); traffic windows span 4/8/16 s. Retrieval returns groups in about two seconds; a cancellable sidecar streams agent rankings. Section 6 gives the benchmark setup.

4 System-1 Retrieval and Group-by-Video Ranking

A heuristic parser routes each query to visual (English translation), OCR (exact and fuzzy Vietnamese phrases), speech, audio and clip channels, and infers folder-scope filters from the topic. Channel c returns a frame ranking L_c, and weighted reciprocal rank fusion [9] scores frame f by $R(f) = \sum_{c:f\in L_c} w_c/(60 + r_c(f))$ with ranks $r_c(f) \geq 1$; cosine scores from different embedding spaces are never added.

Group-by-video ranking. The console groups frames for playback (Fig. 2). For video v, returned frames F_v, best frame $f_v^\star$ and $R^\star = \max_f R(f)$,

$$S(v) = 0.75\,\frac{R(f_v^\star)}{R^\star} + 0.15\,\frac{\text{mean}_{f\in\text{top3}(F_v)}\,R(f)}{R^\star} + 0.10\,\min\!\Big(1, \frac{n_{30}(v)-1}{4}\Big), \quad (1)$$



Fig. 2. Console workflow: query and search controls (left), grouped video results (centre), and frame inspection, evidence and submission identifiers (right).

where $n_{30}(v)$ counts returned frames within 30 s of $f_v^\star$, including itself. These heuristic weights are unablated; top-frame gaps over 30 s flag ambiguity. With TARA, video-level RRF over channels and clip scales replaces Eq. (1).

TRAKE. A shared context (subject, clothing, scene) is added to every event description, events are retrieved separately, temporal non-maximum suppression keeps distinct moments, and an exact dynamic program selects the maximum-weight chain with increasing times in each video. Videos are ranked by coverage, then chain quality.

5 CAD VR: Guarded Evidence Control

Evidence board. Each query owns a fresh board $\mathcal{B}_t$ of constraints, candidates and evidence; no previous hint or reasoning enters it. A deterministic compiler retains the query (weight 2), extracts clauses and attribute, relation or negation spans (weight 1), and adds TRAKE events (weight 1.5) or a QA answer constraint (weight 2). Evidence is de-duplicated by kind and source. An agent's report counts as a visual observation only after that agent loaded the frame; this records inspection, not guaranteed truth. Agent confidences order that agent's proposals but are not used as probabilities.

Typed verification. Each step verifies a shortlist of five candidates, prioritising each agent's leading proposal and the verified leader, then distinct moments from the fused ranking. TRAKE can expand it to 12 to cover the leading video's events. For candidate v and constraint c, Jev 1.13 returns a typed distribution over $\{$*supported*, *refuted*, *unknown*$\}$, with instructions not to infer visual details from embedding scores or confidence. Here "non-generative" describes the decision interface: it returns probabilities rather than free-form plans or search queries.

Algorithm 1 CAD VR controller for one query (6 steps, 40 tool calls, 240 s).

Input: query q, task type, System-1 result R_0

$\mathcal{C}_q \leftarrow$ Compile(q); $\mathcal{B}_0 \leftarrow$ Seed$(R_0, \mathcal{C}_q)$; $\mathcal{I} \leftarrow \emptyset$ ▷ fresh, query-local
for $t = 0, 1, \ldots$ **while** budget remains **do**
  JevVerify$(\mathcal{B}_t)$; Publish(ranking of $\mathcal{B}_t$)
  if guard (3) holds **then return** ▷ evidence sufficient
  $\mathcal{A}_t \leftarrow \{\texttt{STOP}\} \cup \{$calls to agents $\notin \mathcal{I}\} \cup \{$untried evidence actions in mode F$\}$
  $a_t \leftarrow$ JevRoute$(\mathcal{B}_t, \mathcal{A}_t)$; **if** $a_t = \texttt{STOP}$ **then** $a_t \leftarrow$ next escalation in $\mathcal{A}_t$
  if $a_t = \texttt{STOP}$ **then return** ▷ actions exhausted
  $\mathcal{B}_{t+1} \leftarrow$ Execute$(a_t, \mathcal{B}_t)$; $\mathcal{I} \leftarrow \mathcal{I} \cup$ agents in a_t

Without evidence, code sets $p^{\text{sup}} = 0$. It aggregates

$$s(v) = \exp\Big(\sum_{c \in \mathcal{C}(v)} w_c \log \max\big(\epsilon, p_c^{\text{sup}}(v)\big) \Big/ \sum_{c \in \mathcal{C}(v)} w_c \Big), \tag{2}$$

a weighted geometric mean that penalises weakly supported constraints; the separate guard requires every applicable constraint to pass. *Holistic* mode (E) verifies the query and answer constraints; *constraint-level* mode (F) verifies all applicable constraints and enables evidence actions. Ranking promotes candidates for which *supported* is the largest probability on every assessed constraint, demotes those with any *refuted* maximum, and otherwise preserves System-1/agent order. Thus *unknown* is abstention; promotion need not satisfy the stop threshold.

Routing and the stop guard. While the guard fails, Jev chooses the next action from $\mathcal{A}_t \subseteq \{$`STOP`, `CALL_CODEX`, `CALL_CLAUDE`, `CALL_BOTH`, `SEARCH_MORE`, `INSPECT_TOP1`, `COMPARE_TOP2`$\}$, each described in one sentence; the first request also classifies modality and query structure. Each agent runs at most once. The evidence actions (mode F only) are backend macros: a focused search for the least-supported constraint, or OCR and speech around the top two candidates. Agent disagreement enters the state as a signal, and the router is told it alone does not justify another agent. Evidence-sufficient stopping requires, for top candidate v_1,

$$\begin{aligned} &s(v_1) \geq \tau, \qquad s(v_1) - \max_{v' \in \text{comp}(v_1)} s(v') \geq \delta, \\ &p_c^{\text{sup}}(v_1) > \max\big(\tfrac{1}{2},\ p_c^{\text{ref}}(v_1),\ p_c^{\text{unk}}(v_1)\big)\ \forall c, \end{aligned} \tag{3}$$

where competitors exclude the same moment (≤ 8 s apart in one video); TRAKE compares other videos. Only scored competitors count (default zero). QA needs an answer; TRAKE needs supported events in increasing time. We use fixed raw-score thresholds $\tau = 0.9$, $\delta = 0.1$. A rejected `STOP` triggers the next escalation (Codex, Claude, inspect, search, compare). Budget or action exhaustion still returns the current ranking, marked apart from evidence-sufficient stopping (Algorithm 1). Passing the guard is a policy decision, not a correctness guarantee.

System-2 agents and tools. Codex CLI with GPT-6.1 Sol [33] and Claude Code with Opus 5.5 [2] use high reasoning effort and fast mode in fresh processes

without shell/file tools. Both share ten MCP tools for retrieval, corpus browsing, local frame/text inspection, candidate comparison and reporting (inventory in the repository, Sect. 6). Soft roles favour exploration or counterexample inspection without restricting capabilities. Prompts describe the corpus and include a bounded board snapshot marked as untrusted data.

6 Experimental Setup

Benchmark. We assembled 111 queries from the organisers' AIC 2026 query workbooks and final-round submissions. Ground truth combines DRES-accepted submissions [35] with owner-reviewed annotations, each checked against the source frames or clips: moments for all tasks, accepted answers for QA and ordered per-event alternatives for TRAKE. Queries are split by content hash and grouped by target video, so no target video appears in both DEV (25) and TEST (86). TEST has 59 T-KIS, 19 QA and 8 TRAKE queries over 87 target videos, using InfoShot++ (46 queries) or organiser (40) profiles. DEV fixed the policy; its fitted calibration was not adopted, so all configurations use raw probabilities.

Configurations. The initial six share System-1, the soft roles and the ten tools. **A**: retrieval only. **B**: always Codex. **C**: Codex and Claude in parallel after retrieval. **D**: Jev holistic reranking without agents. **E**: CAD VR with holistic verification. **F**: CAD VR with constraint-level verification and evidence actions. B and C fuse retrieval and agent proposals by RRF. System-1 uses PE-Core for visual search on both profiles, heuristic parsing with VI→EN translation, no LLM rewriting or reranking, and the top 150 frames in at most 30 videos; routing activated OCR in 18, speech in 11, audio in 9 and TARA in 5 TEST queries.

Metrics. Strict R@k and MRR require a moment within ± 1 s; QA additionally requires a normalised accepted answer, and TRAKE a complete ordered sequence in one video. *Video* R@k checks target identity among the first k distinct videos. *Group-moment* R@k checks whether those videos contain a returned matching moment, ignoring QA answers but retaining TRAKE's complete-sequence requirement. Video order follows first occurrence in the saved ranking. Rescue@1 measures strict success with agent provenance among runs whose own retrieval seed failed. We record latency, calls and tokens, plus ECE (10 bins) and Brier score [6] over final scored candidates of non-TRAKE queries. Reliability plots use five bins; candidate observations are not independent query samples.

Protocol. Each query–configuration pair is a fresh live run, with variants in seeded random order per query. The initial comparison retains 516 completed TEST runs. The follow-up C/C+V/E round uses the same 86 queries (258 runs), raw policy, prompts, tools, retrieval and budgets. C+V adds E's final holistic verifier to C, with no routing or adaptive stopping, and saves pre-verification rankings. This separate round reuses analysed TEST data without new DEV tuning. Quota-interrupted attempts are retried from fresh state; in the follow-up round, 13

Table 1. Initial TEST round (N=86; tolerance 1 s). R@1: bootstrap 95% CI, 2,000 draws; Resc.: Rescue@1, 73 eligible runs/variant; p50: median latency; Agents: invocations/query; kTok: mean agent and decision tokens in thousands.

	Configuration	R@1 [95% CI]	R@5	MRR	Resc.	p50 (s)	Agents	kTok
A	Retrieval only	15.1 [8.1, 23.3]	33.7	.233	–	1.6	0	0
D	Jev rerank	15.1 [8.1, 23.3]	33.7	.233	0.0	2.1	0	3
B	Always Codex	39.5 [29.1, 50.0]	68.6	.529	28.8	50.5	1.00	230
C	Always both	51.2 [40.7, 61.6]	**72.1**	.612	42.5	51.2	2.00	323
E	CAD VR, holistic	**58.1** [47.7, 68.6]	67.4	**.630**	**52.1**	72.6	1.95	316
F	CAD VR, constraint	52.3 [41.9, 62.8]	64.0	.572	45.2	84.2	1.99	416

interruptions add 314.6 s and 25 agent calls outside the reported completed-run latency, and completed runs have no errors or fallback. Percentile bootstrap intervals [13] resample base queries: 2,000 draws for initial per-configuration intervals and 10,000 paired draws (seed 2026) for contrasts. Exact McNemar tests [30] complement these exploratory, multiplicity-uncorrected comparisons.

7 Results

7.1 Initial Six-Configuration Comparison

E has the highest observed strict R@1 in Table 1 (58.1%). E improves on B by 18.6 points (95% CI $[9.3, 27.9]$; McNemar $p < 0.001$). Its E–C difference is inconclusive (+7.0, $[-2.3, 16.3]$; 11 E-only and 5 C-only successes, $p = 0.21$). C has the highest R@5 (72.1% vs. 67.4%).

D equals A: all 525 assessments were *unknown*, preserving System-1 order, so verifying retrieval-only evidence brings no gain here; this does not isolate verification after agent inspection. F does not improve on E (-5.8 points, $[-15.1, 3.5]$) and changes granularity and evidence actions. F leads initial TRAKE R@1 (50.0% vs. E's 37.5% and C's 25.0%; eight queries only).

7.2 Live C/C+V/E Ablation

Table 2 tests final verification after always-two acquisition. All 86 C+V traces show both agents, one final verification and no routing. C+V–C and E–C+V remain inconclusive. E–C is positive in this round (unadjusted $p = 0.035$), unlike the initial round, but does not establish a gain over the C+V control.

C/C+V use identical acquisition policies but independent trajectories. E–C+V is a controller-package contrast: scheduling, intermediate verification, board-conditioned context and stopping vary together. By task, C/C+V/E obtain 57.6/61.0/74.6% T-KIS, 26.3/26.3/31.6% QA and 25.0/25.0/0.0% TRAKE R@1. E's unchanged aggregate 58.1% across rounds therefore hides task-level variability, including the loss of its three initial TRAKE successes. Decomposing TRAKE, E ranked the target video first for 6/8 initial versus 3/8 follow-up queries and

Table 2. Follow-up TEST round (N=86); recall in percent. Lower panel: paired R@1 contrasts, 95% bootstrap CIs (10,000 draws), exact unadjusted McNemar p and discordant win/loss counts. "Pre": C+V's own ranking before verification.

	R@1	R@5	MRR	p50 (s)	p95 (s)	Agents	Jev calls
C	47.7	**69.8**	.579	53.7	103.2	2.00	0.00
C+V	50.0	66.3	.581	55.6	98.4	2.00	1.00
E	**58.1**	66.3	**.620**	75.8	198.9	1.95	4.91

Contrast	ΔR@1 (pp) [95% CI]	Win/loss	p
C+V − C	+2.3 [−3.5, 9.3]	5/3	.727
E − C+V	+8.1 [0.0, 16.3]	10/3	.092
E − C	+10.5 [2.3, 18.6]	12/3	.035
C+V − Pre	+3.5 [−1.2, 9.3]	4/1	.375

localised 15/28 versus 6/28 events within ±1 s in its top video, so the loss is mainly video identification. Two queries were never solved, and partially aligned chains receive no strict credit.

What does final verification contribute? On identical C+V candidates and answers, final verification raises R@1 from 40/86 (46.5%) to 43/86 (50.0%): four repairs and one harm. This same-evidence difference is inconclusive (Table 2).

In the initial round, re-sorting final candidates by logged RRF, first-seen time and frame ID reproduces all 344 A/B/C/D rankings; it gives E 51/86 vs. verified 50/86 and F 48/86 vs. 45/86 (95% CIs [−5.8, 2.3] and [−9.3, 1.2] points). This limits reranking claims; earlier verification may affect acquisition.

7.3 Accuracy, Latency and Stopping: Router versus Guard

In the initial round, CAD VR did not save computation (Fig. 3): E invoked 1.95 agents per query, and its median latency (72.6 s; p95 171 s) exceeds C's (51.2 s; p95 94 s). The router sent the investigator first in 83/86 queries; sequential escalation adds agent times. Jev averaged 4.9 calls and 3.2 s/query (Fig. 4a). The follow-up round confirms this pattern (Table 2); E/C+V/C use 327k/344k/346k tokens, a workload proxy rather than complete cost.

After the first agent, the initial E router proposed `STOP` with probability above 0.5 in 57/82 decisions (median 0.73): 31/57 top-1 answers were correct, versus 7/25 otherwise (Fig. 4b). The guard blocked all 57; only nine runs ended evidence-sufficient. Offline E† keeps the ranking, calls and time just before the first router `STOP` (else the final outcome): 1.29 agents/query, 35.9 s median and 52.3% R@1, versus E's 1.95, 72.6 s and 58.1%; this post-hoc TEST replay measures neither live latency nor equivalence to C. To select a stricter router on DEV alone, we replay accepting `STOP` after an agent only if $P(\texttt{STOP}) \geq \theta$, with the smallest $\theta \in \{0.50, 0.55, \ldots, 0.95\}$ preserving E's DEV R@1: $\theta = 0.9$. In both TEST rounds it stops 14/15 runs early with no strict outcome changed, cutting agents to 1.79/1.78 but median latency only to 70.7/73.6 s (from 72.6/75.8);

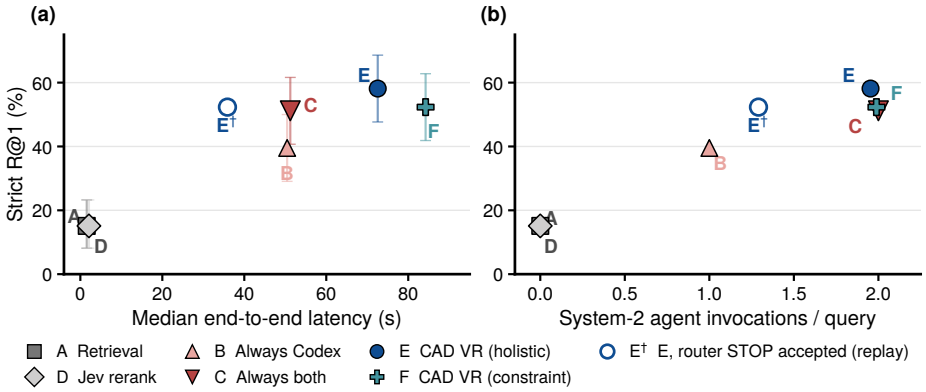


Fig. 3. Initial-round strict R@1 versus latency (a; 95% bootstrap CIs) and agent calls (b). E† is an offline first-`STOP` replay, not a deployed configuration (Sect. 7.3).

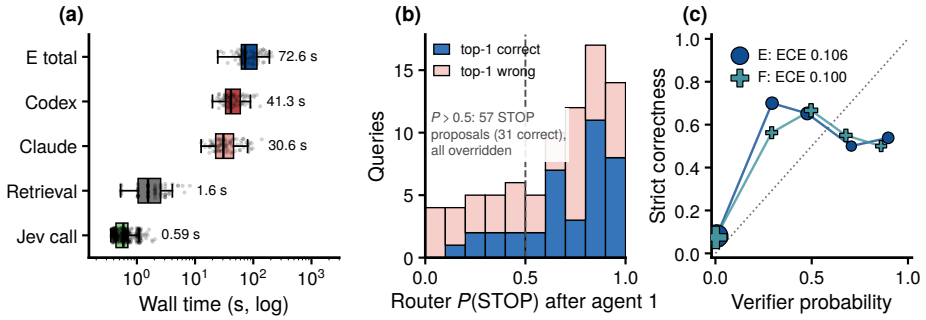


Fig. 4. Initial controller diagnostics. (a) E wall time per layer (log scale). (b) Router P(STOP) after agent 1 by top-1 correctness; every proposal above 0.5 was overridden. (c) E/F verifier reliability (marker area $\propto$ count).

$\theta = 0.7$ gives 51.2% R@1 at 48.1/49.7 s. Sequential escalation dominates latency, motivating a speculative second agent cancelled once the guard passes, to be confirmed live.

Verifier scores imperfectly predict strict correctness (ECE 0.106, Brier 0.112; Fig. 4c): only 7/13 candidates scoring at least 0.8 were correct. Labels concern exact moments/answers, not constraint support.

7.4 Group-by-Video Evaluation

Figure 5 separates exact-answer accuracy from coverage in returned video groups: strict, group-moment and video R@1 are 15.1/43.0/45.3% for retrieval and 58.1/81.4/86.0% for E, and group-moment R@5 is 86–90% with agents. These same-ranking gaps motivate browsing, not a gain over ungrouped retrieval. For QA, E's 89.5% group-moment versus 42.1% strict R@1 mixes exact-frame selection and answer extraction; neither is isolated.

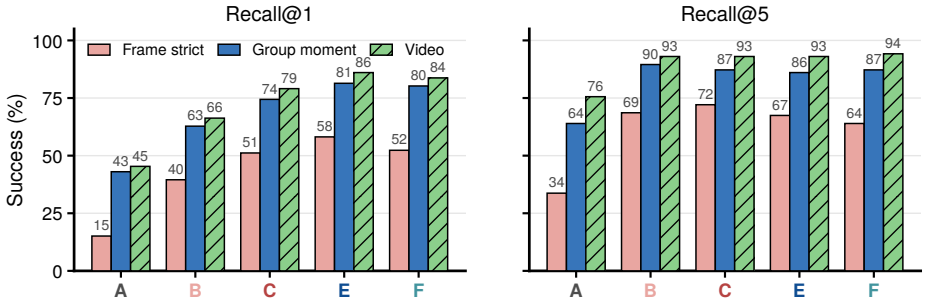


Fig. 5. Initial strict, group-moment and video R@k (D=A; metrics in Sect. 6). This compares metrics, not grouping algorithms.

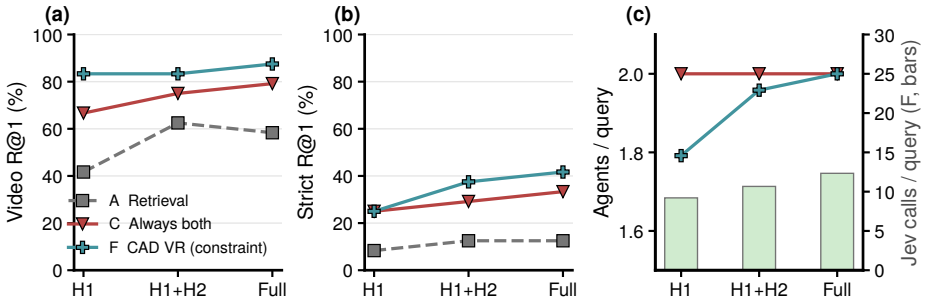


Fig. 6. Cumulative-hint evaluation on 24 TEST queries, each prefix a fresh, independent run. (a) Video R@1. (b) Strict R@1. (c) Agents per query (lines) and Jev calls per query for F (bars).

7.5 Cumulative-Hint Evaluation

Twenty-four TEST queries (12 T-KIS, 12 QA) have three or four reviewed hints whose concatenation equals the full query. Each prefix h_1, $h_1{+}h_2$, … is an *independent* fresh run without cross-hint memory; we compare A, C and F. With only the first hint (Fig. 6a), F ranks the target video first for 83.3% of queries, against 66.7% for C (+16.7 points, $[4.2, 33.3]$) and 41.7% for A. On the full query, strict R@1 is 41.7%, 33.3% and 12.5% (F vs. A +29.2, $[12.5, 50.0]$; F vs. C +8.3, $[-8.3, 25.0]$). F uses 1.79/2.0 agents and 9.2/12.3 Jev calls at first/full hint. Its video R@1 succeeds early on 23/24 queries (Table 3); later prefixes may fail.

Qualitative example. Figure 7 traces an initial-round TEST query through E. Retrieval finds the video but an interview scene. Claude finds the feeding moment; Jev supports it (0.73) and refutes a near-duplicate (0.81). The router proposes `STOP` (0.97), but the guard rejects it ($0.73 < \tau$), triggering Codex's inspection.

Table 3. Hints-to-solve on 24 queries. $h^\star$: first successful prefix (H+1 if unsolved); Early: success before the last hint; Hint saving: $(H - h^\star)/(H - 1)$, 0 if unsolved. These retrospective measures do not imply online stopping or compute savings.

	Video criterion				Strict criterion			
	Solved	Mean $h^\star \downarrow$	Early	Hint saving	Solved	Mean $h^\star \downarrow$	Early	Hint saving
A	19	2.25	.750	.611	6	4.04	.250	.153
C	22	1.67	.875	.792	10	3.42	.375	.319
F	**23**	**1.38**	**.958**	**.889**	**12**	**3.25**	**.458**	**.361**



Fig. 7. Initial-round TEST example. Top: top-1 frames of retrieval (A), always-two (C) and CAD VR (E). Bottom: E's logged controller events (teal) and agent work (amber). The left frame's lower-third is blurred.

8 AIC 2026 Final Round: Results and Practices

Team APCS ⊕ CNTN used this console in the AIC 2026 final round [12]. The organisers scored our T-KIS at 1219.6 (15 queries; Outstanding), QA at 988.2 (12; Outstanding), TRAKE at 57.9 (3; Very Good) and V-KIS, not studied here, at 450.0 (5; Very Good); overall performance was Outstanding. TRAKE is also our least stable benchmark task (Sect. 7.2). Of the 30 final-round T-KIS/QA/TRAKE queries, 26 fall in TEST; there, autonomous E ranks the target video first for 24 and 23 in the two rounds but solves 12 strictly (C: 9; A: 3), with QA at 3/12 strictly versus 11–12/12 by video. The live QA score suggests operators completed this step by localising the frame and reading the answer; their effort was not timed.

Operators searched manually while agents ran on the full query in a separate tab (agents off in the manual tab to avoid cancellation), inspecting streamed candidates and frame/OCR/ASR evidence at once. The final-round cases below are operator observations, not matched ablations or measured speedups; LLM rewriting is outside the benchmark.

T-KIS: select clip retrieval. In `tkis-query-23`, a worker towing a bin by motorcycle drops a white bag. TARA returned the correct segment from hint 1, where PE failed; later hints verified it (`N061-V002`, 289.607 s).

T-KIS: use concurrent agents. In `tkis-query-10`, cyclists pass white balloons onto a bridge before six riders appear frontally. Manual PE/TARA localisation was difficult; a concurrent agent found `S01-V009` at 4542.7 s. Initial A/C rank it second/first; C reports it at 30.4 s, finishing at 62.5 s.

TRAKE: align events manually. The accepted fish-hotpot sequence in `L26_V402` is 91.24, 140.48, 189.12 and 224.24 s. Retain event context and inspect first-contact/last-entry boundaries on paused frames; increasing DP chains cannot guarantee alignment (exact first-frame labels are unavailable), so verify event count, order and identifiers.

9 Discussion and Limitations

E–C+V establishes neither superiority nor equivalence, and scheduling remains unisolated. The DEV-selected stop threshold saves few calls; calibrated stopping needs live evaluation, and TRAKE needs more queries and event-level metrics. Generalisation is limited by corpus size/language, owner-reviewed annotations, thematic overlap across video-disjoint splits, and bootstraps that omit shared-video dependence. Two TEST rounds cannot reliably estimate C/E variability; C+V was run once per query on previously analysed TEST data. Hosted model identifiers cannot freeze behaviour, text-only verification inherits observation errors, and geometric support is uncalibrated for joint correctness.

10 Conclusion

CAD VR makes escalation and stopping auditable; team APCS $\oplus$ CNTN was rated Outstanding in the AIC 2026 final round. E led observed R@1, but gains over C+V were inconclusive and compute savings were not demonstrated. Further live evaluation is needed before causal or efficiency claims.

Acknowledgments. We thank the AIC 2026 organisers and the reviewers. This research is partially supported by the research funding from the Faculty of Information Technology, University of Science, Ho Chi Minh City, Vietnam.

Disclosure of Interests. The authors have no competing interests to declare that are relevant to the content of this article.

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